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Cloud Infrastructure Strategy

Migration of on-premise IT environment to Amazon Web Services

The City of Charles Sturt (CCS) was proclaimed on 1 January 1997. Joining together the former City of Hindmarsh Woodville and the City of Henley and Grange, its vibrant population of 120,000 lives between the city and the sea where the community enjoys a relaxed lifestyle close to the beach and CBD, with easy access to the airport and regional shopping centres. Known as a major sporting and entertainment hub in Adelaide, The City of Charles Sturt has national basketball and soccer stadiums, three golf courses, and an international rowing course. The Adelaide Entertainment Centre hosts world-class music and events, and community centres are constantly hosting a range of workshops and programs.

Comunet has been designing and implementing IT strategies that leverage technology to achieve business objectives for small businesses through to enterprise organisations, for 25 years. With expertise across all areas of IT including strategic consulting, cloud technology, cyber security and software, Comunet helps clients on their digital transformation journey. Through their partnerships with Aamazon Web Services (AWS), they deliver the best technology solutions effectively and efficiently.

The City of Charles Sturt employs over 550 staff and provides residents with Local Government services. The Council's operations are based at the Civic Centre, Woodville. In addition, the Council utilises a works depot located at Beverley. It also operates 5 library branches (Civic, Findon, West Lakes, Hindmarsh and Henley Beach) and several Community Centres.

The challenge

City of Charles Sturt's Information Services (IS) support Council services and corporate business activities, with Information Technology (IT) operations hosted on-premises or cloud based. There are over 80 applications and 1700 IT assets and 50TB data.

With a significant portion of on-premise ICT infrastructure scheduled for replacement, CCS had an opportunity to review how IT services and applications were being hosted. They required a continued focus on risk reduction looking to improve resilience and cyber security posture while reducing operational support risk for its IT services. The council were also were keen to explore innovative technology to support CCS's Smart City initiative allowing the IS team to better respond to its community and business needs.

The analysis

A cloud assessment with business case and TCO modelling was completed by CCS that identified the following business benefits and outcomes for a public cloud migration:

- Improvement in community value through the provision of cloud infrastructure services

- Improvement in the mitigation of risk including addressing single point person risk and improving Disaster recovery practices

- Increased organisational agility through reduction in time to establish new functionality and capacity

- Increased organisational capability by freeing up ICT resources to undertake more value add and less operationally oriented tasks

- Increased capability for innovation.

1 | Design & planning

- Detailed application portfolio & network discovery
- Cloud solution design
- Migration planning

2 | Foundations and migration

- AWS landing zone
- Migration pilot & testing
- Server & application migration phase

3 | Cloud operations, security and governance

- Operational enablement & support
- Security & governance framework
- Skills and training

48 applications migrated to the cloud in **93 days** across **3 portfolios**

The approach

Comunet proposed to meet the requirements of the City of Charles Sturt's public cloud migration by migrating its servers and storage to Amazon Web Services (Sydney Region). AWS are the biggest provider of *Infrastructure as a Service* globally and offer numerous services for networking, server (compute) and storage.

AWS was uniquely positioned to meet CCS's requirements and could be employed to meet mandates, reduce costs, drive efficiencies, and increase innovation. Tens of thousands of government agencies, educational institutions, and non-profits around the world are already using AWS to address a diverse set of client needs, from complex government systems to mission-critical intelligence projects dealing with large volumes of sensitive data.

Comunet undertook a Lift and shift migration based on the AWS Professional Services Migration Acceleration Program (MAP) Methodology. Comunet have trained and worked alongside AWS Professional Services to deliver this program to customers, and now have the experience and capability to deliver a migration methodology that is based on best practices from hundreds of customer migration projects. The Comunet migration programme can significantly reduce time to migrate and lower cost and risk for clients.

"We wanted to better protect our server infrastructure and applications in the event of a disaster.

We wanted also to work with a partner that would allow us to put the brakes on sometimes, allow us to skill our staff up, take the time to instruct them, then fast track again"

Marc O'Conaill

Cloud and Solutions Architect at The City of Charles Sturt

The outcome

Despite a global pandemic hitting midway through the project, Comunet and CCS's IS team were still able to complete the first full data centre migration of infrastructure to AWS by a council in South Australia, on time and budget.

The new environment running in AWS has improved resilience with 99% uptime and a security posture uplift with the implementation of 35 Detective and preventative guardrails and 10 AWS Security services across Identity & access management, logging and monitoring, infrastructure and data protection.

In total the team migrated 48 applications in 93 days across portfolios: Asset management, City Services and Corporate Services from on-premise data centres to AWS.

In October 2020, CCS's IS team won the Local Government IT SA award for Excellence in IT Service Delivery. This award recognises excellence in the delivery of Information Technology services beyond the business as usual; through initiatives that are new to the organisation or new to Local Government.
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